



Synthesis and characterization of poly(2-hydroxyethyl methacrylate/itaconic acid/poly(ethylene glycol) dimethacrylate) hydrogels

Sava N. Dobić, Jovanka M. Filipović, Simonida Lj. Tomić*

Faculty of Technology and Metallurgy, University of Belgrade, Belgrade, Serbia

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ABSTRACT

Hydrogels based on 2-hydroxyethyl methacrylate (HEMA), different poly(ethylene glycol) dimethacrylates (PEGDMA) and itaconic acid (IA) were prepared by free radical crosslinking copolymerization. The influence of different PEGDMA content and type, with PEG chain length of 550 and 875, on the structure, morphology, biocompatibility, swelling and release properties of P(HEMA/IA/550PEGDMA) and P(HEMA/IA/875PEGDMA) copolymeric hydrogels have been investigated. Preliminary biocompatibility characterization of P(HEMA/IA/PEGDMA) hydrogels, done by the cytotoxicity assays using the HeLa cell line, indicated satisfactory cytocompatibility. Swelling studies were conducted for all samples in a physiological pH and temperature range, showing a noticeable pH sensitivity and temperature dependent equilibrium swelling. The characterization by Fourier transform infrared (FTIR) spectroscopy confirmed the presence of all monomers used in the gel structure, and morphology study, performed by scanning electron microscopy (SEM), revealed no significant differences between the samples with respect to pore size. In order to elucidate the release mechanism, drug release study was carried out *in vitro* using two forms of vitamin B₃: nicotinamide (NAM) and nicotinic acid (NAC). The dissolution profiles were fitted to various mathematical models and the best fit was obtained for first order kinetics. The favorable swelling and drug release properties of P(HEMA/IA/PEGDMA) hydrogels make them good candidates to be used as biomaterials.

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1. Introduction

Hydrogels are hydrophilic, weakly crosslinked polymeric biomaterials which can absorb and retain large amounts of water and biological fluids [1–3]. The success of hydrogels originates from their well-known biocompatibility mainly due to their high water content and soft nature. They are important materials in medicine and pharmacy for a variety of applications such as: drug delivery, tissue engineering and in biomedical, food and pharmaceutical industries [3,4].

The homo- and copolymer hydrogels of 2-hydroxyethyl methacrylate (HEMA) have found extensive applications in the field of biomedicine because of their good chemical stability, high biocompatibility and physicochemical properties similar to those of living tissues [5,6].

Different properties of poly(2-hydroxyethyl methacrylate/itaconic acid) (P(HEMA/IA)) hydrogels were investigated in the last decade by few authors. Caykara et al. determined the solubility parameters of pure PHEMA and P(HEMA/IA) hydrogels in 20 solvents, with various solubility parameters, in swelling

experiments [7]. Tomic et al. investigated the swelling behavior and network parameters of hydrogels based on HEMA/IA and different types of poly(alkylene glycol) (meth)acrylates. The same authors [8] investigated swelling and thermodynamic properties of PHEMA and copolymeric P(HEMA/IA) hydrogels with varying IA content in a wide pH and temperature range. These authors also investigated the drug release profiles from P(HEMA/IA) hydrogels in order to investigate transport phenomena. Metal ion adsorption on P(HEMA/IA) hydrogels [9,10] was investigated for uranyl ions by Inam et al. and Ozyurek et al. [11,12]. Hamdy et al. studied the potential use of P(HEMA/IA) hydrogels in immobilization of *Citrullus vulgaris urease*. Immobilized *urease* maintained a higher relative activity than free *urease* at both lower and higher pH levels [13]. Cytotoxicity test and hemolytic activity were investigated by Tomic et al. [14] for poly((alkylene glycol)(meth)acrylates/HEMA/IA) hydrogels. The *in vitro* study of biocompatibility showed no evidence of cell toxicity or any considerable hemolytic activity. All hydrogels showed satisfactory bioadhesive properties.

Poly(ethylene glycol) (PEG) is one of the most widely used synthetic materials for biomedical applications. Its excellent properties make it ideal for use in drug delivery applications, as well as cell delivery vehicles for promoting tissue regeneration [15–18]. Poly(ethylene glycol) dimethacrylate (PEGDMA)

* Corresponding author. Tel.: +381 11 3303 810; fax: +381 11 3370 387.

E-mail address: simonida@tmf.bg.ac.rs (S.Lj. Tomić).

monomers, obtained by substituting PEG terminal hydroxyl groups with methacrylates, are versatile building-blocks for the preparation of “smart” biomaterials. Hydrogels containing PEGDMA crosslinker are highly tunable [19–23].

Several research groups investigated HEMA/PEGDMA hydrogels. Haxhinasto et al. [24] investigated membranes based on HEMA and PEGDMA copolymerized with two different acidic monomers to study cellular attachment and proliferation over the range of prepared charges. Cellular attachment and proliferation increased with increasing proportions of charged monomers. Bako et al. [25] used HEMA and PEGDMA as monomers to prepare nanoparticles and a matrix gel nanocomposite (NCHG), in order to study the release of chlorhexidine as antibacterial drug. The most rapid release was observed over a 24 h period for matrix gels with low crosslinking density. For NCHGs, the release period exceeded 48 h. PEGDMA and PEGDMA-co-hydroxyethyl methacrylate hydrogels were synthesized by Magin et al. [26] to investigate initial attachment of zoospores, cells and the bacteria. PEGDMA-co-HEMA surfaces reduced attachment of spores of *Ulva linza* by 97% compared to a smooth PDMS standard. Bodugoz et al. [27] prepared hydrogel copolymers of HEMA and methacrylic acid with crosslinking agents containing poly(ethylene glycol) (PEG) to study their glucose recognition capabilities. They reported on the structure, diffusive, and recognition characteristics of these gels, the effect of the type and ratio of crosslinking agent, as well as the template/comonomer ratios on glucose binding ability.

In our study copolymeric hydrogels based on 2-hydroxyethyl methacrylate (HEMA), itaconic acid and different types of poly(ethylene glycol) dimethacrylates (P(HEMA/IA/PEGDMA)), which have not yet been investigated, were prepared by free radical crosslinking copolymerization. PEG was chosen to improve the properties of P(HEMA/IA/PEGDMA) hydrogels because of its flexibility, non-toxicity and biocompatibility. Cytocompatibility of the obtained hydrogels was tested using HeLa cells. Fourier transform infrared (FTIR) spectroscopic analysis was used to determine the chemical structure, while scanning electron microscopy (SEM) was applied to investigate the microstructure of hydrogels. The obtained hydrogels were characterized by swelling studies in the physiological pH and temperature range. In order to evaluate P(HEMA/IA/PEGDMA) hydrogels as potential drug carriers in controlled release systems the drug release was studied *in vitro* using vitamin B₃ (nicotinamide and nicotinic acid forms). The experimental data for nicotinamide and nicotinic acid release from P(HEMA/IA/PEGDMA) samples were fitted to the well-known mathematical kinetics models, such as the Higuchi, First order, Peppas and Weibull.

2. Experimental

2.1. Materials

2-Hydroxyethyl methacrylate (HEMA) (Aldrich), freshly distilled, itaconic acid (IA, Fluka), and poly(ethylene glycol) dimethacrylates (PEGDMA, M_n 550 and 875, Aldrich), as received, were the monomers and polymers used in this study. Ethylene glycol dimethacrylate (EGDMA, Aldrich), as crosslinking agent, potassium persulfate (KPS, Fluka), as initiator, and N,N,N',N'-tetramethylethylenediamine (TEMED, Aldrich), as activator, were used in all polymerizations performed in a mixture of water/ethanol as solvent. Two forms of vitamin B₃ were used as drugs: nicotinic acid (NAC, Fluka) and nicotinamide (NAM, Fluka). Potassium hydrogen phosphate (KH₂PO₄ and K₂HPO₄, Fluka) were used for buffer preparation. Demineralized water was used for all copolymerizations and the preparation of the buffer solutions. Scheme 1

2.2. Synthesis of hydrogels

The P(HEMA/IA/PEGDMA) copolymeric hydrogels were prepared by free radical copolymerization and crosslinking. The monomers used in all experiments were HEMA and IA and PEGDMA: the PEGDMA mole fractions were 5, 10 and 15; poly(ethylene glycol) dimethacrylate (M_n 550) was used in the first series and poly(ethylene glycol) dimethacrylate (M_n 875) was used in the second series. The feed compositions for the hydrogels are in Table 1. According to the monomer mole fraction, samples were designated as P(HEMA/IA/550PEGDMA-5), P(HEMA/IA/550PEGDMA-10), P(HEMA/IA/550PEGDMA-15), P(HEMA/IA/875PEGDMA-5), P(HEMA/IA/875PEGDMA-10) and P(HEMA/IA/875PEGDMA-15), respectively. All reactants were dissolved in water/ethanol mixture (Table 1), the reaction mixture degassed and placed between two glass plates sealed with a rubber spacer (2 mm thick). The reaction was carried out at 50 °C for 24 h. After the reaction, the gels were cut into discs and immersed in water for a week to remove unreacted chemicals, by changing water daily. The discs were dried to xerogels (1 mm thick and 5 mm in diameter) in vacuum at room temperature. The amount of unreacted IA was determined by titration of extract against NaOH (0.05 mol/L) to phenolphthalein end point. On the other hand, the amount of unreacted HEMA and PEGDMA were determined using a UV spectroscopy. In both cases, results indicated that the yields of various compositions were all above 99%. The dry gels were transparent flexible discs with a limited fluid absorption capability. The dimensions of the xerogel samples were: thickness of 0.115 ± 0.001 cm, diameter of 0.526 ± 0.001 cm, and weight of 0.0256 ± 0.0001 g (in average) for all measurements.

2.3. Cytotoxicity test

One type of human cells (human cervix epithelial carcinoma cells (HeLa)) was used for the cytocompatibility study. The xerogel samples were sterilized by UV radiation. Thereafter, the hydrogel samples were immersed in ultra purified water for 24 h. During this period, the water was refreshed every several hours. For the cytotoxicity test, 200 μ L of HeLa cells in DMEM, with a concentration of 5.0×10^4 cells/mL was firstly added to each well in a 24-well plate and incubated for 24 h in an incubator (37 °C, 5% CO₂). Subsequently, the culture medium was changed to 200 μ L of DMEM containing the hydrogel sample with particular weight and incubated for another 48 h. Then DMEM with the hydrogel was substituted by fresh DMEM, and 30 μ L of MTT solution (5 mg/mL) was added to the HeLa cells. After incubation for 4 h, 200 μ L of DMSO was added and the mixture was shaken at room temperature. Cell viability (%) was calculated as follows (Eq. (1)):

$$\text{cell viability (\%)} = \left(\frac{\text{number of viable cells}}{\text{total number of cells}} \right) \times 100 \quad (1)$$

2.4. FTIR spectral analysis

Xerogels were crushed into a powder and mixed with potassium bromide (Merck IR spectroscopy grade) in the proportion 1:100, and dried at 40 °C. The mixture was compressed to a 12 mm semi-transparent disk by applying a pressure of 65 kN (Pressure gauge, Shimadzu) for 2 min. FTIR spectra over the wavelength range 4000–700 cm^{−1}, with a resolution of 4 cm^{−1}, were recorded using a FTIR spectrometer (BOMEM Michelfan MB-102 FTIR).

2.5. Scanning electron microscopy (SEM)

A scanning electron microscope (JEOL JSM-6460 LV) was used to observe specimen morphologies of xerogels. The lyophilized

Feed compositions for the P(HEMA/IA/PEGDMA) hydrogels (same conditions for 550PEGDMA and 875PEGDMA component).

Component	P(HEMA/IA/PEGDMA-5)	P(HEMA/IA/PEGDMA-10)	P(HEMA/IA/PEGDMA-15)
HEMA (mol%)	90	85	80
IA (mol%)	5	5	5
PEGDMA (mol%)	5	10	15
HEMA + IA + PEGDMA (wt%)	10	10	10
Demineralized water (wt%)	45	45	45
Ethyl alcohol (wt%)	45	45	45

Scheme 1. Chemical structures of components used for hydrogel synthesis.

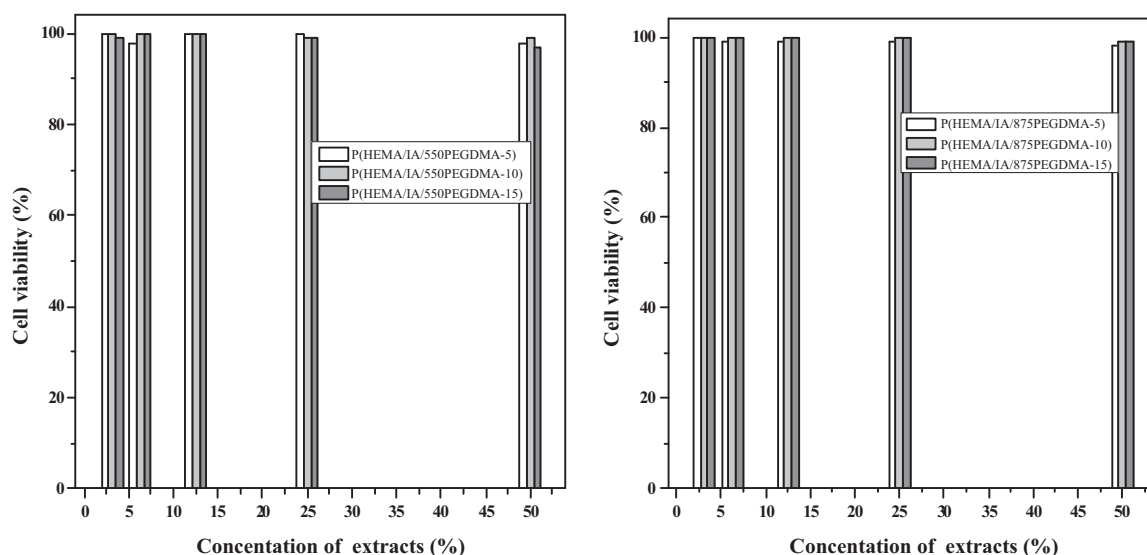


Fig. 1. Cell viability of the P(HEMA/IA/550PEGDMA), and P(HEMA/IA/875PEGDMA) hydrogels in the cytotoxicity assay.

3.2. Spectral characteristics of P(HEMA/IA/PEGDMA)

FTIR spectroscopic analysis was used to illustrate the chemical structure of the hydrogels and the nature of the bonds formed. All characteristic bands which are present in FTIR spectra of components ((monomers HEMA and IA), and polymer (PEGDMA)) appear into the FTIR spectrum of the obtained hydrogel (Fig. 2). The spectral characteristics of P(HEMA/IA/PEGDMA), presented in Fig. 2, revealed the characteristic stretching vibration band of hydrogen-bonded alcohol (O–H) around 3450 cm^{-1} , the C=O stretching vibration of the ester group also appeared at 1730 cm^{-1} , and an absorption band with a weak shoulder around 2950 cm^{-1} , which correspond to the stretching of aliphatic $-\text{CH}_2-$, C–H and $-\text{CH}_3$ groups, respectively (Fig. 2). The increased peak intensity of the C=O group at 1730 cm^{-1} in the spectrum of P(HEMA/IA/550PEGDMA-15) gel was associated with the presence of the additional C=O groups from IA. On the other hand, several bands appeared in the fingerprint region for ethylene glycol units, originating from PEGDMA component, between 1600 and 1000 cm^{-1} . These peaks were assigned to the $-\text{CH}_2$ scissoring band of ethylene

glycol units at 1480 cm^{-1} and the antisymmetric and symmetric stretching bands ($-\text{O}-\text{R}$) of ethylene glycol units at 1160 cm^{-1} , respectively. Other characteristic bands represent C–C and C–H vibrations of $-\text{CH}_3$ and $-\text{CH}_2-$ groups [31]. Therefore, the FTIR spectroscopy results confirmed the incorporation of poly(ethylene glycol) dimethacrylates and itaconic acid in these HEMA based-hydrogels.

3.3. Morphology of P(HEMA/IA/PEGDMA)

Surface morphology of the resultant hydrogels were examined under SEM. Fig. 3 represents the scanning electron micrographs of P(HEMA/IA/550PEGDMA-10) and P(HEMA/IA/550PEGDMA-15) samples. The lyophilized P(HEMA/IA/PEGDMA) samples have a porous structure, with honey-comb like texture. The pore sizes of samples are in the nonporous range showing no significant differences between the samples, which is consistent with the swelling results.

3.4. Swelling studies and network parameters

It is worth to investigate the swelling behavior of hydrogels with respect to the environmental stimuli which are very important factors for their application. The swelling studies were performed to investigate the influence of the external stimuli (pH and temperature) and hydrogel composition on the dynamic and equilibrium swelling properties of P(HEMA/IA/PEGDMA) hydrogels [32]. The pH and temperature dependences of equilibrium swelling degree (q_e) for P(HEMA/IA/PEGDMA) hydrogels are presented in Figs. 4 and 5, respectively, in a pH physiological range of 2.2–8.0, at 37°C and temperature range of 25 – 55°C , in a buffer of pH 7.40.

pH-sensitive P(HEMA/IA/PEGDMA) hydrogels show a sharp volume phase transition at pHs near pK_a values for IA, i.e. they have very different equilibrium swelling degrees in various physiological media. This pH-sensitive support can protect the stability of drug (at low pH) and release the drug at the targeted site (at medium or high pH), depending upon the composition of the hydrogel and the drug in each drug delivery system. The swelling studies reveal the anionic character of P(HEMA/IA/PEGDMA) hydrogels, due to the presence of IA carboxyl groups with $pK_{a1} = 3.85$ and $pK_{a2} = 5.45$, which are responsible for the pH-sensitive swelling behavior. The q_e vs pH dependences (Fig. 4) have the same trend for all the samples: at pH values lower than both pK_{a1} and pK_{a2} values, the swelling

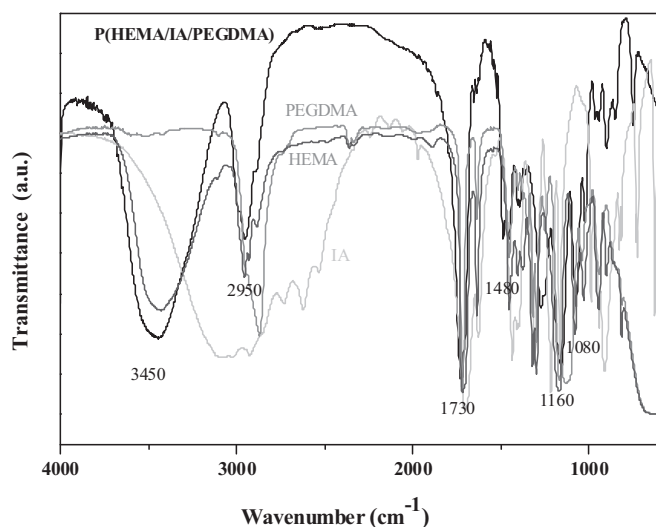


Fig. 2. FTIR spectra of components: HEMA, IA, PEGDMA, and P(HEMA/IA/550PEGDMA-15) gel.

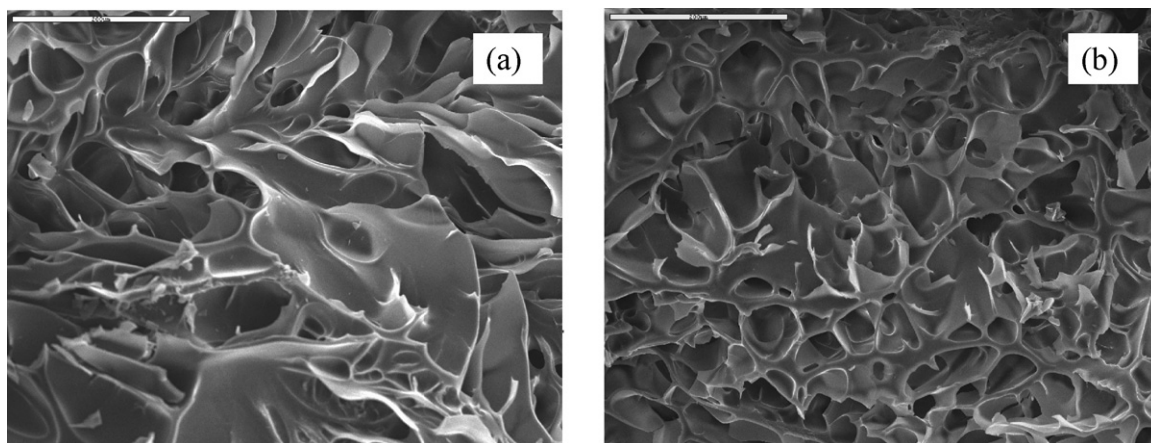


Fig. 3. SEM micrographs of resultant hydrogels (a) P(HEMA/IA/PEGDMA550-10) and (b) P(HEMA/IA/550PEGDMA-15) ('bar' 200 μm, 200×).

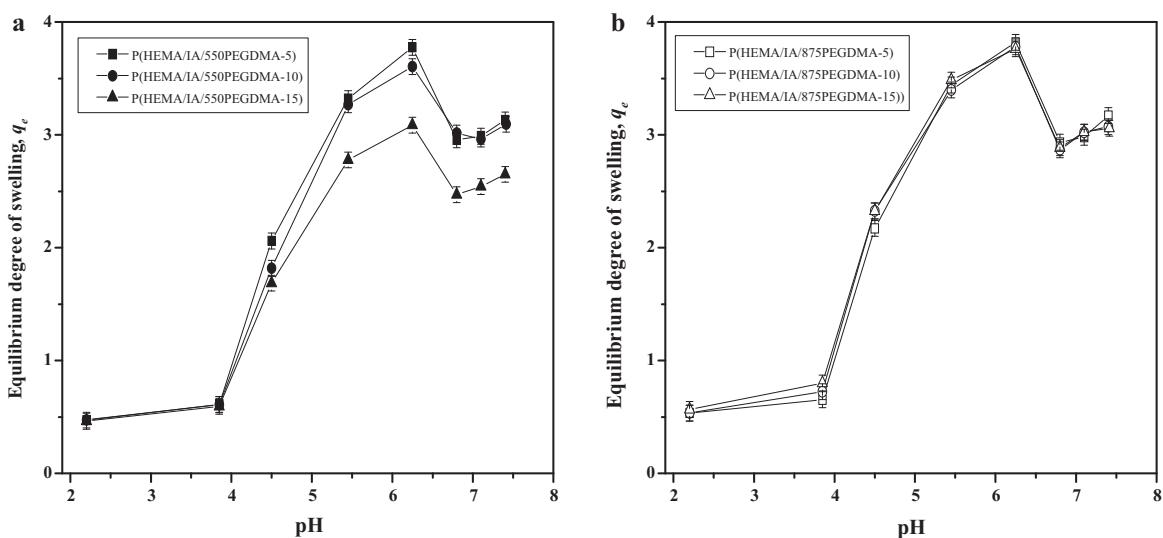


Fig. 4. pH sensitive swelling behavior of P(HEMA/IA/550PEGDMA) (a) and P(HEMA/IA/875PEGDMA) (b) hydrogels at 37 °C.

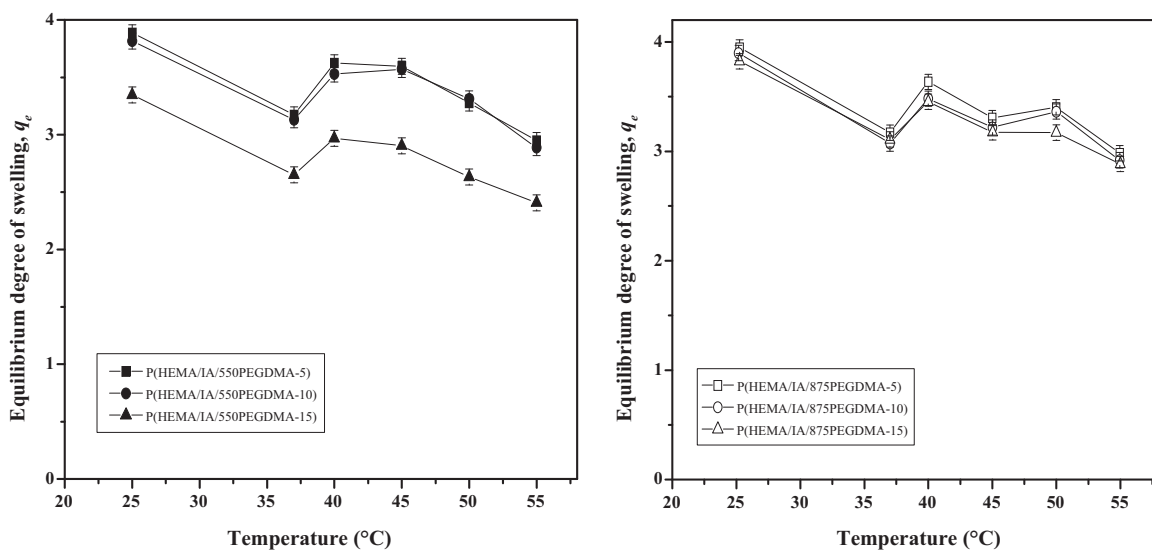


Fig. 5. q_e vs temperature dependences of P(HEMA/IA/550PEGDMA) (a) and P(HEMA/IA/875PEGDMA) (b) hydrogels in a buffer pH of 7.40.

degrees are low and slightly depend on the composition of the samples, i.e. the swelling degree is low and almost similar for all the samples. Low q_e values are primary due to the intermolecular physical crosslinks, via hydrogen bond formation, between carboxylic groups in IA with the hydroxyl groups of HEMA or with ether groups in PEGDMA residues. q_e values drastically increase with the increase of pH up to pH 6.2, which is the maximum value, than they slightly decrease (in the pH range of 6.2–6.8) and finally increase (in the pH range of 6.8–8.0). As the pH value of the surrounding medium raises to 6.20, which is above pK_a values of both carboxyl groups, those groups are transformed to carboxylate anions, the hydrogen bonds are broken and the hydrogel swelling is much higher. In the pH range of 4.54–6.80 the P(HEMA/IA/550PEGDMA) samples show the dependence of equilibrium swelling degrees on 550PEGDMA content (Fig. 4(a)) whereas such a dependence was not noticeable for P(HEMA/IA/875PEGDMA) samples (Fig. 4(b)).

The samples with 550PEGDMA show different swelling profiles: 550PEGDMA-5 has the highest q_e values, while the sample 550PEGDMA-10 swells a little less and 550PEGDMA-15 has the lowest q_e values. This difference in swelling capacity with the change of 550PEGDMA content can probably be explained by the following argument. By introducing PEGDMA chains in the hydrogel structure two opposing effects take place: PEGDMA chains are hydrophilic and highly elastic, so they will induce higher swelling, but at the same time they act as crosslinking agent, which will reduce the swelling degree. For the samples containing shorter 550PEGDMA chains there is a critical concentration of 15 mol% where the crosslinking effect outweighs the chain flexibility effect. The longer 875PEGDMA chains are more flexible and the crosslinking effect is balanced with the chain flexibility effect, so the swelling behavior is similar for all samples in the range of concentrations used in this investigation.

q_e vs temperature dependence shows a similar trend for all the samples (Fig. 5(a) and (b)). Generally, q_e values decrease from 25 to 37 °C, then slightly increase from 37 to 40 °C, and finally decrease in the interval from 40 to 55 °C. The equilibrium swelling degrees decrease with the rise of temperature above 40 °C because at higher temperatures the extra thermal energy provided leads to thermal dissociation of hydrogen bonds between water and corresponding functional groups in hydrogel which favors hydrophobic interaction and the hydrogel chains swell less. q_e vs temperature dependences for P(HEMA/IA/550PEGDMA) samples depend on 550PEGDMA content, whereas for P(HEMA/IA/875PEGDMA) samples there was practically no difference in q_e with varying 875PEGDMA content, which trend is the same as in the case of q_e vs pH dependences (Fig. 4(a) and (b)). This can be explained by the fact that longer 875PEGDMA chains are more flexible to accommodate to the temperature and pH changes than shorter 550PEGDMA chains. It is obvious that P(HEMA/IA/PEGDMA) hydrogels show temperature dependent swelling, but are not temperature sensitive because there is not a drastic change of q_e at any temperature in the measured range. It is known that hydrogels based on PHEMA or its copolymers with IA also have no LCST volume phase transitions [33].

Itaconic acid in P(HEMA/IA/PEGDMA) hydrogels is responsible for pH sensitive behavior and the hydrophilic/hydrophobic interactions govern the temperature dependent behavior. Therefore, the change of environmental conditions tunes hydrogel swelling. Besides, the q_e values for P(HEMA/IA/PEGDMA) were substantially higher and the temperature dependent swelling more pronounced than in the case of P(HEMA/IA) samples [33], which is due to hydrophilic and flexible PEG chains.

Properties of hydrogels with diprotic acid moieties depend on the network structure, which is controlled by the feed composition and the parameters of the reaction. The most important network parameters are the average molar mass of the polymer chain

between two neighboring crosslinking points (M_c), the effective crosslinking density (ν_e) and pore size (ξ). The molecular weight between the crosslinks, for ionic hydrogels is calculated using the Eq. (3) [34]:

$$\frac{V_1 X^2 \phi_{2s}^2}{4I \bar{V}_r^2} \left(\frac{2K_{a1}K_{a2} + 10^{-\text{pH}}K_{a1}}{2(10^{-\text{pH}})^2 + 10^{-\text{pH}}K_{a1} + K_{a1}K_{a2}} \right)^2 = [\ln(1 - \phi_{2,s}) + \phi_{2,s} + \chi \phi_{2,s}^2] + \left(\frac{V_1 \rho}{\bar{M}_c} \right) \phi_{2,r}^{2/3} \phi_{2,s}^{1/3} \quad (3)$$

where K_{a1} and K_{a2} are the first and second dissociation constants of a diprotic acid, X is the weight fraction of ionizable polymer in the system, I is ionic strength of the swelling medium, $\phi_{2,s}$ is the polymer volume fraction in the swollen gel, $\phi_{2,r}$ is the polymer volume fraction in the relaxed state, V_1 is the molar volume of water, ρ is the polymer density, $\bar{V}_r$ is the average molar volume of polymer repeating units and χ is the Flory polymer–solvent interaction parameter. The effective crosslinking density (ν_e) was calculated as $\nu_e = \rho/M_c$.

The network pore size (ξ) represents the distance between consecutive crosslinking points, which describes the available space for solute transport within the polymer network, was calculated according to Eq. (4), which is described in more detail by Canal and Peppas [35]:

$$\xi = \phi_{2,s}^{-1/3} l \left(\frac{2C_n \bar{M}_c}{M_r} \right)^{1/2} \quad (4)$$

Here M_r is the average of the molar mass of the repeating units, l the C–C bond length (1.54×10^{-10} m for vinyl polymers), and C_n is the Flory characteristic ratio [36].

According to the physiologically interesting values for the potential biomedical application in dermatology and drug delivery systems, the calculations were done for swelling kinetic data obtained at pH of 7.40 and 37 °C.

The results for network parameters of P(HEMA/IA/PEGDMA) hydrogels are presented in Table 2. Small differences in network parameter values are due to different PEGDMA fractions and PEG chain lengths (550 and 875) in gels, as the concentrations of other reactants in the samples are constant. The values of effective crosslinking densities (ν_e) are in the range of 0.113–0.196 mol/dm³. Pore size values of P(HEMA/IA/PEGDMA) hydrogels, are in non-porous range. The polymer–solvent interaction (χ) parameter values for the hydrogels are around 0.6, which values are lower than those for PHEMA and P(HEMA/IA) hydrogels, revealing that P(HEMA/IA/PEGDMA) hydrogels are more hydrophilic. The gel structure and properties are correlated as it can be seen from pore size results which correlate well with swelling results (Fig. 4, and Table 2).

3.5. Fitting the *in vitro* drug release data

The release of drug molecules from a hydrogel depends on many factors, such as the chemical structure of the polymer, the network structure, the release conditions, etc. On the other hand, the chemical structure and the molecular weight of the drug also influence the release behavior. The present study focused on *in vitro* release of vitamin B₃, which comes in two basic forms – niacin (also called nicotinic acid) and niacinamide (also called nicotinamide), from P(HEMA/IA/PEGDMA) hydrogels.

Based on the swelling results, four samples were chosen to evaluate the vitamin B₃ release profiles, since the drug release properties of hydrogels can be connected with their swelling properties. Therefore, the samples P(HEMA/IA/550PEGDMA-5) and P(HEMA/IA/550PEGDMA-15) were chosen for *in vitro* drug release experiments due to the

Table 2

Network parameters of P(HEMA/IA/PEGDMA) hydrogels in a pH of 7.40, at 37 °C.

Sample	q_e	χ	M_c (g/mol)	ν_e (mol/dm ³)	ξ (nm)
P(HEMA/IA/550PEGDMA-5)	3.17	0.605	7634.1	0.113	6.420
P(HEMA/IA/550PEGDMA-10)	3.13	0.606	7545.5	0.117	5.638
P(HEMA/IA/550PEGDMA-15)	2.65	0.625	4715.6	0.196	4.468
P(HEMA/IA/875PEGDMA-5)	3.17	0.605	8159.8	0.113	6.305
P(HEMA/IA/875PEGDMA-10)	3.07	0.608	7752.4	0.120	5.449
P(HEMA/IA/875PEGDMA-15)	3.03	0.605	6392.7	0.125	4.509

most pronounced differences in swelling behavior. The sample P(HEMA/IA/550PEGDMA-10) was omitted because its q_e value is very similar to that of P(HEMA/IA/550PEGDMA-5). The samples P(HEMA/IA/875PEGDMA-5) and P(HEMA/IA/875PEGDMA-15) were chosen to compare samples of different PEGDMA molecular weights. Figs. 6 and 7 represent the drug release profiles of chosen hydrogel samples in a buffer pH of 7.40, at 37 °C.

From Fig. 6(a) it can be observed that the drug release profiles for NAm release depend slightly on the 550PEGDMA content in the copolymers, while this difference is more pronounced for NAc release (Fig. 6(b)) which is consistent with their swelling behavior

(Fig. 4(a)). The release profiles for NAm have a burst effect and the drug release is faster, whereas the release profiles of NAc do not show the burst effect and the release rates are slower than in the case of NAm.

The release profiles of NAc and NAm from 875PEGDMA samples (Fig. 7(a) and (b)) reveal that the release rates are higher in the case of NAm than for NAc, and have a burst effect but the release profiles do not differ much for different PEGDMA contents. The higher release rates for both drugs in the case of 875PEGDMA are due to longer, more flexible PEG chains which make the system more dynamic.

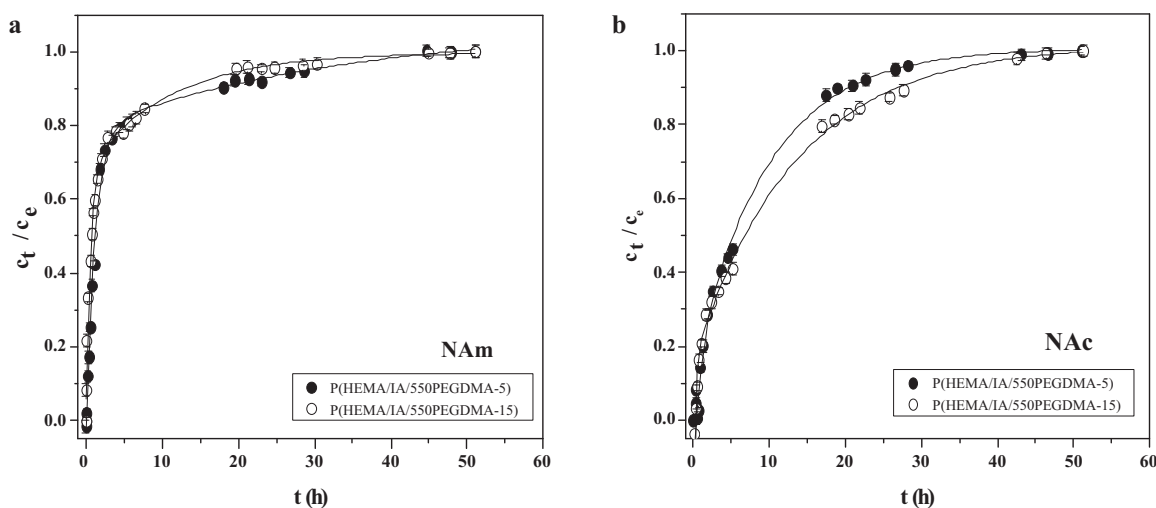


Fig. 6. Release profiles of nicotinamide (NAm) (a) and nicotinic acid (NAC) (b) from P(HEMA/IA/550PEGDMA) hydrogels in *in vitro* conditions.

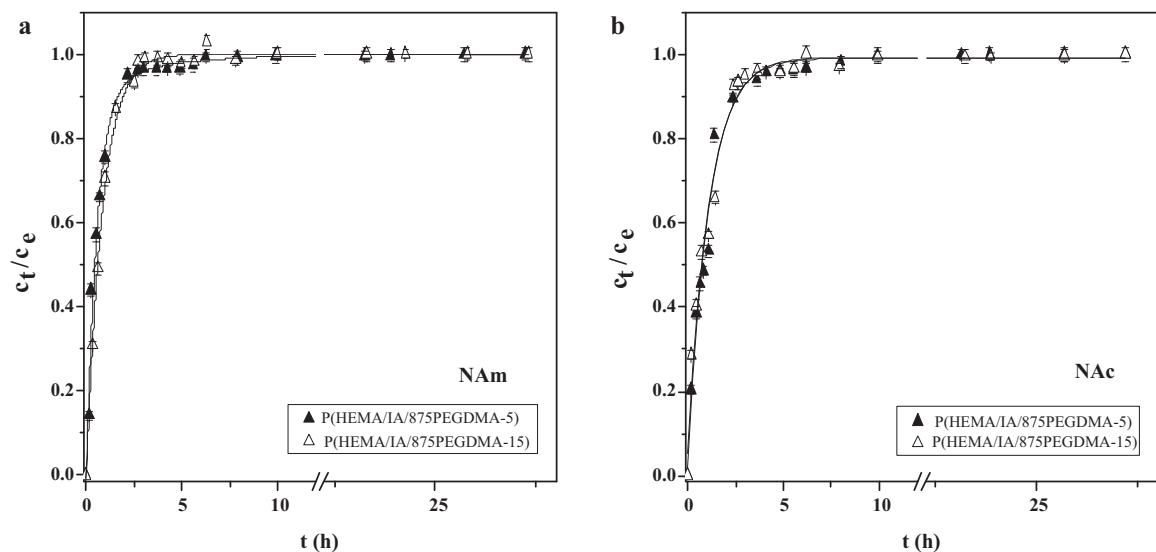


Fig. 7. Release profiles of nicotinamide (NAm) (a) and nicotinic acid (NAC) (b) from P(HEMA/IA/875PEGDMA) hydrogels in *in vitro* conditions.

Table 3Release rate coefficients for nicotinamide *in vitro* release data fitted with different release models (all the release experiments were performed in triplicate).

Release model		550-5	550-15	875-5	875-15
Higuchi	k/SD^a	0.652/0.001 ^a	0.732/0.001	0.857/0.001	0.901/0.001
	r^2	0.3397	0.3461	0.4035	0.5336
Peppas	k/SD^a	0.914/0.001	0.911/0.001	0.705/0.001	0.744/0.001
	n	0.594	0.871	0.142	0.121
	r^2	0.9790	0.8290	0.6309	0.5201
First order	k/SD^a	0.594/0.001	0.871/0.001	1.62/0.01	1.18/0.01
	r^2	0.9797	0.9297	0.9853	0.9958
Weibull	k/SD^a	0.576/0.001	0.008/0.001	0.038/0.001	4.145/0.001
	n	0.872	0.206	0.212	1.178
	r^2	0.9742	0.9224	0.6359	0.9927

^a Standard deviation of k for 3 repeated experiments.**Table 4**Release rate coefficients for nicotinic acid *in vitro* release data fitted with different release models (all the release experiments were performed in triplicate).

Release model		550-5	550-15	875-5	875-15
Higuchi	k/SD^a	0.458/0.001 ^a	0.434/0.001	0.814/0.001	0.838/0.001
	r^2	0.24395	0.3220	0.34509	0.41667
Peppas	k/SD^a	1.034/0.001	0.984/0.001	0.636/0.001	0.665/0.001
	n	0.0877	0.0793	0.174	0.156
	r^2	0.9491	0.9442	0.6788	0.6637
First order	k/SD^a	0.0877/0.0001	0.0793/0.0001	0.925/0.001	0.956/0.001
	r^2	0.9496	0.9448	0.9842	0.9827
Weibull	k/SD^a	0.0733/0.0001	0.0664/0.0001	0.781/0.001	0.665/0.001
	n	0.547	0.623	1.373	1.510
	r^2	0.8577	0.8942	0.9779	0.9786

^a Standard deviation of k for 3 repeated experiments.

The following model equations (Eqs. (5)–(8)) describing drug release were used for fitting the release data using linear and non-linear regression analysis:

$$\text{First order equation } F = 1 - e^{-kt} \quad (5)$$

$$\text{Higuchi equation } F = kt^{1/2} \quad (6)$$

$$\text{Peppas equation } F = kt^n \quad (7)$$

$$\text{Weibull equation } F = 1 - e^{-kt^n} \quad (8)$$

(F = fractional drug release; k = release rate constant for the different equations; n = diffusional exponent)

Different mathematical models (Eqs. (5)–(8)) were applied to drug release data for two forms of vitamin B₃ from PEGDMA-5 and PEGDMA-15 samples, with different PEGDMA molecular weights, in order to evaluate the release kinetics and mechanism. The criteria for selecting the most appropriate model were based on linearity (coefficient of correlation). The results of fitting experimental data to mathematical models are presented in Tables 3 and 4, respectively, indicated that drug release followed the first order kinetics. The pharmaceutical dosage forms which follow the first order kinetics release the drug in a way that is proportional to the amount of drug remaining in its interior, in such a way, that the amount of drug released by unit of time was diminished. The drug release mechanism for these two drugs is by diffusion.

4. Conclusions

A new type of crosslinked anionic polymers, containing HEMA, IA, and PEGDMA components, were designed and synthesized to be tested as matrices for controlled drug release application. The effects of composition, and environmental conditions on P(HEMA/IA/PEGDMA) hydrogel properties were investigated. The results of *in vitro* studies of the cytotoxicity showed that they had no apparent cytotoxicity. FTIR analysis showed that the hydrogels had

the expected chemical composition. The pore sizes of samples were in the nonporous range with no significant differences between the samples. Swelling studies showed pH-sensitive behavior, typical for anionic hydrogels, and temperature dependent swelling. Due to their pH sensitive swelling behavior P(HEMA/IA/PEGDMA) copolymers can be considered as smart hydrogels. The swelling results also revealed that PEGDMA residues improved hydrogel swelling capabilities and enhanced their temperature dependence. The release profiles for two forms of vitamin B₃ indicate that the initial release rate of drugs, NAc and NAM, could be controlled by modifying the composition of the matrix. The *in vitro* vitamin B₃ release results were fitted to well-known kinetic models, such as Higuchi, First order, Peppas and Weibull, in order to evaluate the best fit for P(HEMA/IA/PEGDMA) hydrogels as matrices for the release of two forms of vitamin B₃, nicotinamide and nicotinic acid. The vitamin B₃ release data fit well the first order kinetic law for all the samples. These pH-responsive and temperature dependent hydrogels could be beneficial for the use as drug delivery systems.

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